

Virtual Business Manager for Indian Artisans

Smart India Hackathon 2026 — Technical Proposal

Problem Domain	Digital Commerce & AI for Marginalized Communities
Track	Empowering Artisans, Weavers & Micro-Entrepreneurs
Platform	Cross-platform Mobile App + WhatsApp Bot
Primary Stack	Flutter · FastAPI · PostgreSQL · Bhashini · Claude AI
Target Users	Artisans, Weavers, Handicraft Makers — Low Digital Literacy
Integration	GeM Portal · ONDC · Bhashini · GI Registry

"From village craft to digital marketplace — in under 5 minutes, in your own language."

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1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

India has over 7 million registered artisans, weavers, and handicraft makers. The government actively supports them through financial assistance, cluster development, and exposure via physical fairs such as Shilp Samagam, Surajkund Mela, and Dilli Haat. However, these events offer only temporary income boosts. Between fairs, artisans have no reliable digital sales channel — not because the markets don't exist, but because the barrier to entering digital commerce is too high.

Pain Point	Root Cause	Impact
Cannot photograph products professionally	No studio, no skills, no equipment	Listings look uncompetitive
Cannot write product descriptions	Low English literacy, no copywriting knowledge	Poor SEO, low discoverability
Cannot price competitively	No access to market benchmarks	Either undersell or lose buyers
Cannot navigate GeM/ONDC portals	Complex UI, English-only, multi-step process	Exclusion from govt marketplaces
Cannot reach B2B buyers year-round	Dependent on physical fair schedules	Feast-or-famine income cycle

ShilpSetu Solution:

A unified AI-powered platform with two entry points — a Flutter mobile app for power users and a WhatsApp bot for field use — both backed by the same backend. An artisan onboards once via WhatsApp OTP, then seamlessly moves between both interfaces. The AI handles photography, description, pricing, GI tagging, marketplace submission, and buyer matching — all in regional languages, with voice as the primary input method.

2. UNIFIED PLATFORM ARCHITECTURE

2.1 The Dual-Interface Philosophy

ShilpSetu is not two separate products. It is one backend serving two client interfaces optimised for different contexts. The WhatsApp bot is the field tool — used at fairs, in workshops, on the road. The mobile app is the power dashboard — used for analytics, bulk management, and GeM submissions. One login, one account, one inventory.

Client Layer	Flutter App (Android/iOS) + WhatsApp Bot (Twilio Sandbox / Meta Cloud API)	Flutter 3.x · Dart
API Gateway	Single FastAPI backend serving both clients via REST + WebSocket	FastAPI · Uvicorn · Nginx
Auth Service	WhatsApp OTP login · JWT tokens · Phone-number-based identity	Twilio Verify · PyJWT
AI Orchestrator	Routes tasks to appropriate AI service — image, NLP, pricing, GI, story	LangChain · Python
Image Service	Background removal, enhancement, multi-background mockups	rembg · Pillow · Cloudinary
Voice/NLP Service	Speech-to-text, translation, description generation, TTS reply	Bhashini API · Claude API
Pricing Service	Market scraping, ML price prediction, GeM/Amazon/Flipkart comparison	SerpAPI · scikit-learn
GI Service	GI tag matching from artisan region + craft type	Static GI DB · fuzzy matching
Matchmaking	Vector similarity between product embeddings and buyer preferences	pgvector · OpenAI Embeddings
Data Store	Primary DB · Vector store · Cache · File storage	PostgreSQL · pgvector · Redis · S3
Integrations	GeM portal · ONDC network · Bhashini · GI Registry	GeM API · ONDC SDK
Infra	Containerised deployment · CI/CD · Auto-scaling	Docker · Railway/Render · GitHub Actions

2.2 Data Flow — WhatsApp to Live Listing

1	Artisan sends photo + voice ▼	WhatsApp message hits Twilio webhook → FastAPI receives multipart payload
2	Auth check ▼	Phone number looked up in users table → JWT context established for this session
3	Media extraction ▼	Photo saved to S3 temp bucket · Voice OGG extracted and sent to Bhashini STT

4	AI pipeline ▼	Parallel: image enhancement (rembg) + description generation (Claude) + price fetch (SerpAPI)
5	GI check ▼	Region + craft matched against GI database → badge assigned if qualifying
6	Listing created ▼	Product record written to PostgreSQL · Storefront URL generated · Embeddings stored in pgvector
7	Bot reply ▼	WhatsApp bot sends back: listing preview text + Bhashini TTS audio + storefront link
8	App sync	Flutter app reflects new product instantly via WebSocket push

3. CORE FEATURE 1 — AI IMAGE STUDIO

Professional product photography is the single biggest barrier to competitive e-commerce listings. Artisans photograph products on floors, in cluttered workshops, with bad lighting. The AI Image Studio transforms any photo into a marketplace-ready image automatically.

3.1 Technical Implementation

Input	Raw JPEG/PNG from WhatsApp (compressed) or in-app camera (full resolution)
Step 1: Upscale	Real-ESRGAN model upscales compressed WhatsApp images 4x before processing
Step 2: Segment	Meta's SAM (Segment Anything Model) or rembg (U2-Net) isolates product from background
Step 3: Enhance	Pillow + OpenCV: auto white-balance, contrast normalization, sharpening
Step 4: Backgrounds	3 variants generated: (a) Pure white/Amazon-style (b) Lifestyle mockup (c) Festive seasonal
Step 5: QC Gate	CLIP model scores image quality — rejects blurry/dark/occluded photos, asks for retake
Output	3 professional JPEGs uploaded to Cloudinary CDN with optimised URLs for mobile delivery
Library Stack	rembg 2.x · Real-ESRGAN (rrdbnet) · Pillow 10.x · OpenCV 4.x · Cloudinary Python SDK
Hosting	GPU inference on Replicate.com (free tier: 500 predictions/month) or Modal.com
WhatsApp Workaround	In-app camera shortcut shared via WhatsApp — captures full-res before WhatsApp compression

3.2 Background Generation Details

Once the product is segmented, three background composites are generated using pre-designed templates. White background uses a flat 255,255,255 canvas with a soft drop shadow. Lifestyle mockups use curated background plates (e.g., wooden shelf for pottery, fabric draped surface for textiles) with the product composited using perspective transform. Seasonal backgrounds (Diwali, Eid, Christmas) are swapped automatically based on calendar proximity to boost festive sales.

Hackathon Demo Tip: Use Replicate API for rembg inference — no GPU server needed. Free tier covers all demo traffic. Fall back to rembg local CPU if Replicate is down.

4. CORE FEATURE 2 — MULTILINGUAL VOICE CATALOGER

Voice is the natural input modality for low-literacy users. The cataloger accepts voice notes in any of India's 22 scheduled languages, converts them to structured product data, and generates SEO-optimised listings in English and Hindi automatically.

4.1 Bhashini Integration Architecture

API	Bhashini ULCA Platform — Government of India, MeitY (free for developers)
ASR Model	AI4Bharat IndicConformer — supports Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Odia, Punjabi + 12 more
Input Format	WhatsApp voice note (OGG Opus) → converted to WAV 16kHz mono → Bhashini ASR REST API
Translation	Bhashini NMT (Neural Machine Translation) — regional language → English
NLP Extraction	Claude claude-sonnet-4-6 extracts structured attributes: material, color, size, technique, region, craft type
Description Gen	Claude generates 3 variants: marketplace title (60 chars) · product card (150 chars) · SEO description (400 chars)
TTS Reply	Bhashini TTS converts confirmation message back to artisan's language — sent as WhatsApp voice note
Endpoint	https://dhruva-api.bhashini.gov.in/services/inference/pipeline
Auth	Bhashini API key (free registration at bhashini.gov.in)
Fallback	If Bhashini ASR fails: OpenAI Whisper-large-v3 via API (paid fallback, ~\$0.006/min)

4.2 NLP Extraction Prompt Design

Claude receives the translated text and extracts structured JSON:

```
SYSTEM: You are a product cataloger for Indian handicrafts. Extract structured attributes.
Return ONLY valid JSON with keys: craft_type, material, color[], size, technique, region,
gi_candidate (bool), title_en, title_hi, description_en, description_hi, tags[]. INPUT: "Yeh
Madhubani painting hai, kagaz pe bani hai, 12x18 inch, Mithila Bihar se, natural rang use kiye
hain."
```

4.3 SEO Optimisation

Generated descriptions are automatically optimised for GeM portal search, Amazon India search, and Google Shopping. Keywords are injected from a craft-specific keyword database (e.g., 'handmade', 'GI certified', 'tribal art',

'eco-friendly', craft-specific regional terms). Title follows Amazon's title formula: Brand + Material + Craft Type + Size + Color.

5. CORE FEATURE 3 — DYNAMIC PRICING ENGINE

Artisans consistently underprice their work because they have no visibility into what similar products sell for online. The pricing engine scrapes live market data from Flipkart, Amazon, Meesho, and GeM, applies a machine learning model, and suggests an optimal price range with reasoning in the artisan's language.

5.1 Market Data Sources & Scraping Strategy

Flipkart	SerpAPI Google Shopping results filtered to flipkart.com domain · Extracts: price, rating, reviews, seller type
Amazon India	SerpAPI Amazon endpoint (amazon.in) · Extracts: price range, BSR rank, Prime vs non-Prime pricing
Meesho	SerpAPI Google Shopping · Meesho typically shows lowest market price — used as floor price
GeM Portal	GeM Product Search API (open, no auth required) · Government buyer reference price
Indiamart	SerpAPI web search for wholesale/bulk price signals
Tool	SerpAPI (100 free searches/month on trial · \$50/month for 5,000 searches at scale)
Query Strategy	Search query auto-built from NLP extraction: '{craft_type} {material} {size} handmade' across all platforms
Data Points	Min price · Max price · Median price · Top-rated seller price · GI-tagged premium (if applicable)
Refresh Rate	Prices cached in Redis for 24 hours per (craft_type, material, size) combination

5.2 ML Price Prediction Model

A scikit-learn Random Forest Regressor is trained on a dataset of 50,000+ handicraft listings scraped from Amazon, Flipkart, and GeM. Features include craft type (one-hot encoded), material, size (numeric), region, GI tag status, image quality score (from CLIP), and description quality score. The model outputs a predicted fair price, which is blended 60/40 with the live market median.

1	Scrape ▼	SerpAPI fetches top 20 results for the product query across all platforms simultaneously
2	Parse ▼	Price parser extracts numeric values, filters outliers (>2 std dev from median removed)

3	Segment ▼	Results split into: mass-market (Meesho/unbranded) · mid-market · premium/GI
4	ML Score ▼	Random Forest model predicts price from product features
5	Blend ▼	Final price = 0.6 * market_median + 0.4 * ml_prediction
6	GI Uplift ▼	If GI tag detected: apply +20-40% premium based on craft-specific historical data
7	Explain ▼	Claude generates 2-sentence Hindi explanation: 'Flipkart pe aisi cheez ■450-600 mein bik rahi hai...'
8	Output	Price range (min/max) + recommended price + 1-line reasoning sent to artisan via bot/app

5.3 Weekly Mandi Intelligence (Automated Push)

Every Monday at 8am, a cron job re-runs price scraping for all active listings. If a product's current price is more than 15% below the new market median, the bot automatically messages the artisan in their language with a price-raise suggestion and one-tap update button.

6. CORE FEATURE 4 — ONE-TIME WHATSAPP LOGIN & ACCOUNT SYNC

The artisan authenticates exactly once — via WhatsApp OTP. From that moment, their account is accessible from both the WhatsApp bot and the Flutter app with the same phone number. No passwords, no email, no separate signup flows.

Auth Method	Phone number as primary identity · OTP delivered via Twilio Verify (WhatsApp OTP or SMS fallback)
Token	JWT (HS256) · 30-day expiry · Refresh token stored in Redis · Invalidated on new login
Bot Session	Phone number mapped to user_id in PostgreSQL · Each webhook call re-validates JWT from session store
App Login	'Login with WhatsApp' button → enters phone number → OTP sent → JWT stored in Flutter SecureStorage
Sync Mechanism	WebSocket connection from Flutter app · PostgreSQL LISTEN/NOTIFY triggers push on data change
Conflict Resolution	Last-write-wins with server timestamp · Optimistic locking on product edits
Offline App	Flutter Drift (SQLite) stores local copy · Background sync via WorkManager on connectivity restore
Security	Rate limiting: 5 OTP requests per phone per hour · Webhook signature validation (Twilio X-Twilio-Signature)

6.1 Onboarding Conversation Flow (WhatsApp)

1	Scan QR ▼	Artisan scans QR at fair/exhibition → WhatsApp opens pre-filled with join message
2	Language Select ▼	Bot sends: 'Namaste! Apni bhasha chunein: 1-Hindi 2-Bengali 3-Tamil 4-Telugu 5-Marathi...'
3	Name & Craft ▼	Bot collects name and primary craft type via 2 conversational messages in chosen language
4	OTP Verify ▼	Twilio Verify sends OTP via WhatsApp to same number → artisan replies with 6-digit code
5	Profile Complete ▼	Account created · Bot confirms: 'Aapka account ban gaya! Pehla product add karein? Photo bhejiye.'
6	App Link (Optional)	Bot sends app download link with deep-link that auto-logs in via same phone JWT

7. CHERRY FEATURE 1 — GI TAG AUTO-DETECTOR (Asli Pehchaan)

India has over 400 registered Geographical Indication (GI) tags for handicrafts, textiles, and food products. Most artisans are unaware that their product qualifies for a GI tag — which can command a 20-40% price premium and significantly improve trust with buyers. ShilpSetu detects GI eligibility automatically and adds a verified badge to the listing.

Data Source	GI Registry database from CGPDTM (Controller General of Patents, Designs & Trade Marks) — public data
Database	Static JSON/PostgreSQL table: {gi_name, craft_type[], state, district[], description, registration_no}
Matching Logic	Fuzzy string match (RapidFuzz library) on craft_type from NLP extraction vs GI craft_type[] · Region match on state/district from user profile
Confidence Threshold	Match score > 85% triggers GI candidate flag · Claude verifies with 1-shot prompt
Badge	Gold 'GI Certified' badge added to storefront · Listing title prefixed with '[GI Tag]' on GeM
Price Impact	Pricing engine automatically applies GI premium on price suggestion
User Communication	Bot message in artisan's language: 'Aapka {craft} {state} ka GI tag ke liye eligible hai! Yeh aapke product ki value badhata hai.'
Future	Connect to CGPDTM API when available for real-time verification · GI certificate upload for full verification

8. CHERRY FEATURE 2 — KAHAANI: CULTURAL STORY GENERATOR

Premium buyers and urban consumers don't just purchase a product — they purchase a story, a heritage, a connection to craft traditions. Kahaani automatically generates a rich cultural narrative for each product, dramatically increasing perceived value and emotional connection with buyers.

Trigger	Automatically generated for every new listing · Can be regenerated on request
Knowledge Base	Pre-compiled JSON corpus of 500+ Indian crafts: history, technique, cultural significance, key producing regions, artisan communities · Sourced from INTACH, Ministry of Textiles, Wikipedia
Retrieval	Vector search (pgvector) on craft knowledge base using product embedding as query · Top 3 relevant chunks retrieved
Generation	Claude claudesonnet-4-6 with RAG: 'Given this craft knowledge and this specific artisan's details, write a 100-word product story in English and Hindi that a premium e-commerce buyer would find compelling.'
Output	English story (100 words) + Hindi story (100 words) + 3 social media caption variants
Display	'The Story' section on artisan's storefront page · Collapsible in marketplace listing
Differentiator	Each story is personalised with artisan's name, village, and years of experience — not generic craft description

Sample Output (Warli Art, Palghar, Maharashtra):

"This Warli painting emerges from the hands of Savita Dhodi, a third-generation tribal artist from Palghar district of Maharashtra. For over 2,500 years, Warli artists have used rice paste on mud-plastered walls to document the rhythms of harvest, marriage, and community. Savita's geometric language — circles, triangles, squares — is not decoration. It is a living scripture, passed mother to daughter through generations. When you bring this painting home, you preserve a voice."

9. CHERRY FEATURE 3 — SHILPI SCORE: SELLER TRUST RATING

B2B buyers placing bulk orders need trust signals before committing. The Shilpi Score is a publicly visible composite score (0-100) displayed on every artisan's storefront, designed to give buyers confidence and motivate artisans to improve their listing quality.

Score Component	Weight	How Measured	Max Points
Listing Quality	25%	Image quality (CLIP score) + description length + attribute completeness	25
GI Tag Status	15%	Verified GI tag present on at least one listing	15
Order Fulfillment	20%	% of orders fulfilled on time (from buyer confirmations)	20
Response Time	10%	Avg time to reply to buyer inquiries via bot/app	10
Buyer Ratings	20%	Average star rating from completed orders (1-5 scale)	20
Profile Completeness	10%	Photo, bio, village, craft type, years of experience all filled	10

10. CHERRY FEATURE 4 — MANDI INTELLIGENCE: WEEKLY PRICE PULSE

Every Monday morning, ShilpSetu sends every active artisan a personalised market intelligence report via WhatsApp — in their language — showing what similar products sold for this week across major platforms. This is financial data that was previously only available to large sellers with analytics teams.

Trigger	Cron job — every Monday 08:00 IST · Celery Beat scheduler on backend
Process	For each active listing: re-run SerpAPI price scrape → compare to artisan's current price → calculate delta
Threshold	If artisan's price is >15% below market median → flag for price-raise suggestion
Message Format	WhatsApp text + optional voice (Bhashini TTS): Platform prices, artisan's price, recommendation
One-Tap Update	Message includes: 'Price update karna chahte hain? Reply 1 for YES' → Auto-updates listing price on confirmation
Trend Arrows	Delta vs last week shown with emoji: rising (green up arrow), falling (red down arrow), stable (equals sign)
Analytics	Admin dashboard shows aggregate pricing trends per craft category — useful for government policy insights
Tech	Celery + Redis (task queue) · SerpAPI batch queries · Twilio bulk message API

11. CHERRY FEATURE 5 — B2B BUYER MATCHMAKING (Udyog Mitra)

The platform maintains a buyer-side registry of B2B buyers — export houses, boutique brands, NGOs, government procurement officers, hotel chains — who have registered their sourcing interests. When an artisan lists a new product, vector similarity matching connects them with the most relevant buyers automatically.

Buyer Registry	Pre-seeded for demo with 50 mock buyers · Real platform: buyer self-registration portal with category preferences
Buyer Profile	{buyer_id, name, org, craft_interests[], min_order_qty, max_budget, preferred_regions[], contact}
Product Embedding	OpenAI text-embedding-3-small on product title + description + tags → 1536-dim vector stored in pgvector
Buyer Embedding	Same embedding model on buyer's interest description → stored at registration time
Matching	pgvector cosine similarity query: SELECT * FROM buyers ORDER BY embedding <=> \$product_embedding LIMIT 5
Threshold	Only buyers with similarity score > 0.78 are shown · Prevents irrelevant matches
Notification	Artisan gets WhatsApp message: '3 buyers interested in your Madhubani painting! View → [link]'
Buyer Side	Buyer gets email/WhatsApp digest of new matching products weekly
Privacy	Buyer contact shown to artisan only after mutual interest confirmation — prevents spam
Stack	pgvector 0.6 · OpenAI Embeddings API · PostgreSQL 16

12. CHERRY FEATURE 6 — GeM ONE-TAP SUBMISSION

The Government e-Marketplace (GeM) is India's official government procurement portal. Listing on GeM gives artisans access to government buyers — the largest procurement system in India with over Rs. 2 lakh crore in annual transactions. The current process requires artisans to navigate a complex multi-step English portal. ShilpSetu reduces this to a single tap.

Eligibility Gate	Shilpi Score > 65 · At least 1 listing with GI tag or verified craft type · GSTIN or Udyam registration
GeM Integration	GeM Seller API (sandbox available for development) · OAuth2 flow links artisan's GeM account to ShilpSetu
Auto-Fill	Product data from ShilpSetu DB auto-populates GeM listing fields: title, description, HSN code, price, images
HSN Code	Auto-assigned from craft_type using HSN code lookup table (Chapter 46, 57, 58, 63 for handicrafts/textiles)
Compliance Check	Pre-submission validator checks GeM's mandatory fields · Flags missing certifications before submission
Status Tracking	GeM approval status polled every 6 hours → bot notifies artisan when listing goes live
ONDC Integration	Parallel submission to ONDC network via Seller App SDK → product visible on all ONDC buyer apps (Meesho, Paytm, etc.)
Demo Approach	GeM sandbox API for demo · Show the auto-filled form + submission flow as screen recording if API unavailable

13. CHERRY FEATURE 7 — MELA MODE: OFFLINE-FIRST SYNC

Physical fairs like Dilli Haat and Surajkund Mela — specifically mentioned in the problem statement — often have poor cellular connectivity due to crowd density. Mela Mode lets artisans photograph and queue all their products offline. The moment they get connectivity, everything uploads and gets AI-processed automatically in the background.

Detection	Flutter ConnectivityPlus plugin detects network state · Mela Mode banner auto-appears on weak/no connection
Local Storage	Flutter Drift (SQLite wrapper) stores: product photos (compressed), voice recordings, metadata, queue status
Queue	Each pending product is a queue entry with status: PENDING / UPLOADING / PROCESSING / DONE / FAILED
Background Sync	Flutter WorkManager triggers sync job when connectivity restored · Runs even if app is in background
Upload Order	Smallest files first to maximise successful uploads on weak connections · Resumes from checkpoint on failure
AI Processing	Server processes queue items asynchronously · Celery tasks with priority queue for Mela Mode uploads
Progress	Real-time progress shown in app: 'Uploading 4 of 7 products... 3 queued'
WhatsApp Fallback	If app sync fails: artisan can WhatsApp photos later and bot links them to their existing account
Storage Limit	Local queue stores up to 200 products offline · Old completed items auto-cleared to free space

14. CHERRY FEATURE 8 — SAHAAYAK: AI NEGOTIATION COACH

Negotiation is where artisans lose the most money. When a buyer haggles, artisans typically cave immediately out of fear of losing the sale. Sahaayak is an AI coach embedded in the WhatsApp bot — the artisan forwards a buyer's message, and the AI suggests a smart, confident counter-response in their language.

Trigger	Artisan forwards buyer WhatsApp message to the ShilpSetu bot number · Bot detects forwarded message format
Context Fetch	Bot identifies which product is being discussed from price/description keywords · Fetches product data + market price
Analysis	Claude analyzes: buyer's offer, artisan's listed price, market range, GI status, bulk quantity signals
Response Generation	Claude generates counter-offer script in artisan's language: what to say, what to emphasize, when to offer discount
Tactics	Includes tactical advice: mention GI tag, offer free delivery as sweetener, suggest smaller trial order first
Example Input	Buyer: 'Bhai ₹400 mein 5 piece doge?' → Artisan forwards to bot
Example Output	Bot: 'Kehna: Sir, ek piece ka rate ₹480 hai. 5 piece ke liye ₹2,200 dunga — aur packaging free. GI certified maal hai, quality pakki. Amazon pe yahi ₹550 mein bik raha hai.'
Learning	Artisan can report if the tactic worked → feedback improves future suggestions (fine-tuning dataset)

15. CHERRY FEATURE 9 — VERNACULAR VIDEO REEL GENERATOR

Word-of-mouth is the primary marketing channel for artisans today. ShilpSetu weaponizes it digitally by auto-generating a 15-second product reel — ready to share on WhatsApp Status, Instagram Reels, or Facebook — with regional music, Hindi/regional text overlay, and the artisan's branding.

Trigger	Generated automatically after listing is created · Available in app under 'Share' tab
Input	3 product images (enhanced by AI Studio) + product title + price + artisan name + craft type
Template Engine	5 pre-designed video templates per craft category: textiles, pottery, painting, jewelry, woodwork
Text Overlay	Product name in Hindi (Noto Sans Devanagari font) + price + 'Handmade in [Region]' + artisan name
Music	Royalty-free regional folk music tracks (pre-licensed) · Auto-selected by region: Rajasthani folk, Bengali baul, Tamil carnatic instrumental
Generation	FFmpeg on backend: image slideshow + ken-burns effect + text overlay + music mix · Output: 1080x1920 MP4 (Instagram Reels format)
Duration	15 seconds: 4s per product image + 3s end card with price + QR code linking to storefront
Delivery	Video sent back via WhatsApp + download link · One-tap Share button opens WhatsApp Status composer with video pre-loaded
Hosting	FFmpeg serverless on Modal.com (free tier) · Video stored on Cloudinary with 30-day expiry
Fallback	If video generation fails: auto-generated image carousel (3 images) as fallback shareable

16. FULL TECH STACK & API REFERENCE

Layer	Technology	Purpose	Cost
Mobile App	Flutter 3.x (Dart)	Cross-platform iOS + Android	Free
State Mgmt	Riverpod 2.x	App state management	Free
Local DB	Drift (SQLite)	Offline-first Mela Mode	Free
Background Sync	WorkManager (flutter_workmanager)	Background upload on connectivity restore	Free
API Backend	FastAPI (Python 3.12)	REST API + WebSocket server	Free
Task Queue	Celery + Redis	Async AI processing + cron jobs	Free tier
Primary DB	PostgreSQL 16	All structured data	Free (Railway)
Vector DB	pgvector extension	Embeddings for matchmaking + RAG	Free
Cache	Redis 7	Session store + price cache + queue	Free tier
File Storage	Cloudinary	Image + video CDN delivery	Free (25GB)
WhatsApp	Twilio Sandbox / Meta Cloud API	Bot messaging webhook	Free (sandbox)
OTP Auth	Twilio Verify	Phone OTP verification	Free trial
Voice STT	Bhashini ULCA (AI4Bharat)	Regional language speech-to-text	Free (govt API)
Translation	Bhashini NMT	Regional language to English/Hindi	Free (govt API)
TTS	Bhashini TTS	Text to voice reply in regional lang	Free (govt API)
AI/LLM	Claude claude-sonnet-4-6 (Anthropic)	Description, story, negotiation, pricing explain	\$3/M tokens
Image AI	rembg + Real-ESRGAN (Replicate)	Background removal + upscaling	Free (500/mo)
Price Scraping	SerpAPI	Flipkart, Amazon, GeM price data	Free (100/mo)
Embeddings	OpenAI text-embedding-3-small	Product + buyer vector embeddings	\$0.02/M tokens
Video	FFmpeg (Modal.com serverless)	Product reel generation	Free tier
Deployment	Railway.app / Render.com	Backend hosting with auto-scale	Free tier
CI/CD	GitHub Actions	Automated test + deploy pipeline	Free
Monitoring	Sentry + Logfire	Error tracking + observability	Free tier
GeM	GeM Seller API (sandbox)	Government marketplace submission	Free
ONDC	ONDC Seller App SDK	Open network distribution	Free

17. DATA MODELS & DATABASE SCHEMA

Table: users

Column	Type / Constraint
user_id	UUID PRIMARY KEY
phone_number	VARCHAR(15) UNIQUE NOT NULL
name	VARCHAR(100)
language_pref	VARCHAR(10) DEFAULT 'hi'
village	VARCHAR(100)
district	VARCHAR(100)
state	VARCHAR(50)
craft_types	TEXT[]
gem_seller_id	VARCHAR(50) NULLABLE
shilpi_score	INTEGER DEFAULT 0
created_at	TIMESTAMP DEFAULT NOW()

Table: products

Column	Type / Constraint
product_id	UUID PRIMARY KEY
user_id	UUID FK → users
title_en	VARCHAR(200)
title_hi	VARCHAR(200)
description_en	TEXT
description_hi	TEXT
craft_type	VARCHAR(100)
material	VARCHAR(100)
color	TEXT[]
size_cm	VARCHAR(50)
technique	VARCHAR(100)
gi_tag	VARCHAR(100) NULLABLE
price_suggested	DECIMAL(10,2)
price_listed	DECIMAL(10,2)
image_urls	TEXT[]
video_reel_url	VARCHAR(500) NULLABLE

embedding	vector(1536)
status	ENUM: draft/active/sold/gem_submitted
created_at	TIMESTAMP DEFAULT NOW()

Table: buyers

Column	Type / Constraint
buyer_id	UUID PRIMARY KEY
org_name	VARCHAR(200)
contact_email	VARCHAR(200)
craft_interests	TEXT[]
min_order_qty	INTEGER
max_budget_per_unit	DECIMAL(10,2)
preferred_regions	TEXT[]
embedding	vector(1536)

Table: orders

Column	Type / Constraint
order_id	UUID PRIMARY KEY
product_id	UUID FK → products
buyer_id	UUID FK → buyers
quantity	INTEGER
total_amount	DECIMAL(12,2)
status	ENUM: pending/confirmed/shipped/delivered/cancelled
buyer_rating	INTEGER NULLABLE (1-5)
created_at	TIMESTAMP DEFAULT NOW()

18. WHATSAPP BOT CONVERSATION FSM

The WhatsApp bot is implemented as a Finite State Machine (FSM). Each user session maintains a current state in Redis. Incoming messages trigger state transitions and actions.

State	Trigger / Input	Actions	Next State
IDLE	New photo received	Save to S3 temp · Acknowledge receipt	AWAIT_DESCRIPTION
IDLE	'mera saman' text	Fetch user's active listings · Format list	SHOW_INVENTORY
IDLE	'aaj ki kama'i' text	Fetch orders today · Calculate total	SHOW_EARNINGS
IDLE	'keemat' text	Ask which product to price-check	AWAIT_PRICE_QUERY
IDLE	Forwarded buyer message	Detect forward · Extract buyer offer · Fetch product details	NEGOTIATION_COACH
AWAIT_DESCRIPTION	Voice note received	Bhashini STT → Claude extract → generate description	AWAIT_CONFIRM
AWAIT_DESCRIPTION	Text message received	Use text as description directly → Claude enhance	AWAIT_CONFIRM
AWAIT_CONFIRM	Reply '1' (Haan)	Create product DB record · Generate storefront listing	AWAIT_PRICE_CONFIRM
AWAIT_CONFIRM	Reply '2' (Badlo)	Delete temp data · Ask for new description	AWAIT_DESCRIPTION
AWAIT_PRICE_CONFIRM	Reply '1' (Theek hai)	Set price · Make listing active · Send storefront listing	IDLE + Bhashini TTS confirm
AWAIT_PRICE_CONFIRM	Reply '2' (Price batao)	Ask artisan to type their desired price	AWAIT_CUSTOM_PRICE
AWAIT_CUSTOM_PRICE	Numeric text	Validate price range · Update listing price · Activate listing	IDLE
NEGOTIATION_COACH	Forward received	Claude analyze + generate counter-script in artisan language	IDLE
SHOW_INVENTORY	Reply with product number	Show product details + edit/delist options	PRODUCT_MGMT
PRODUCT_MGMT	Reply 'delist'	Set product status = inactive · Confirm	IDLE

Session state is stored in Redis with a 24-hour TTL. Each state is keyed by phone number. A 'RESET' command or 30-minute inactivity timeout returns any session to IDLE.

19. IMPACT GOALS & SUCCESS METRICS

Impact Goal	Current State	Target (12 months)	Metric
Digital presence	0 artisans per fair have online listing	80% after onboarding	% listing active listing
Listing time	60-90 min to create a marketplace listing	5 min manually per product	Avg time product → live
Price realisation	Artisans price 30-40% below market	Within 10% of market median	Price vs market benchmark
GI tag awareness	< 5% of eligible artisans know about GI	50% of eligible artisans GI-flagged automatically	% automatically identified
GeM listings	Complex process, < 2% of eligible artisans active	20% artisans active GeM with at least 1 GeM listing	% submissions
Annual income	Avg ₹85,000/year for craft artisans	20% AGR increase via digital channels	Income delta (survey)
Digital literacy	Cannot navigate any e-commerce platform	Platform age WhatsApp bot listings independently	% self-serve after 1 week

20. IMPLEMENTATION ROADMAP & TEAM SPLIT

Phase	Hours	Features	Owner
Phase 0 Setup	0-4h	Repo setup · FastAPI skeleton · PostgreSQL schema · Flutter project · Twilio sandbox · Bhashini connected · Bhashini API key	Full team
Phase 1 Core MVP	4-16h	WhatsApp onboarding FSM · Photo receive + rembg background removal · Bhashini TTS + Claude description · Basic product listing	Backend + Claude
Phase 2 Pricing	16-24h	SerpAPI price scraping · Price suggestion logic · Pricing reply to artisan · Redis cache for prices	Backend
Phase 3 App	16-28h	Flutter UI: home, listing grid, add product, earnings · WebSocket sync · Drift offline DB · Beta Mode	Flutter
Phase 4 Cherries	28-40h	GI tag detector · Kahaani story generator · Shilpi Score · B2B matchmaking (pgvector) · Bhashini Shalpyak negotiation coach	Backend + Shalpyak
Phase 5 GeM + Video	40-50h	GeM sandbox submission flow · ONDC SDK integration · FFmpeg video reel generation · Bhashini Mandi Intelligence cron	Full team
Phase 6 Polish	50-60h	UI/UX refinement · Error handling · Demo data seeding · Pitch deck · Video demo recording	Full team

Suggested Team Roles

Role	Responsibilities
Backend Lead (1)	FastAPI server · PostgreSQL schema · Celery tasks · GeM/ONDC integration · Pricing service
AI/ML Engineer (1)	Bhashini integration · Claude API prompts · rembg pipeline · pgvector matchmaking · GI detector
Flutter Developer (1)	Full app UI/UX · Drift offline DB · WebSocket · WorkManager · app store build
Bot Developer (1)	Twilio webhook · FSM implementation · WhatsApp conversation flows · Bhashini TTS replies
Full-Stack / DevOps (1)	Railway deployment · CI/CD · SerpAPI scraping · FFmpeg video service · Redis config
UI/UX + Presenter (1)	Figma designs · Demo script · Pitch deck · Impact data · Video recording

"ShilpSetu meets artisans where they already are — WhatsApp — and grows with them into a full digital business. From a voice note in Bhojpuri to a live GeM listing in under 5 minutes. No app required. No English required. No technical skills required. This is not a feature. This is access."